

Sandro Sage

AI & Machine Learning Engineer

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[LinkedIn](#)



Experience

ARRK Engineering GmbH

AI Engineer

Munich, Germany
December 2025 – present

- Serve as the leading AI Engineer on a large-scale BMW project; trusted to assume internal project coordination responsibilities when the project lead is unavailable
- Design and develop end-to-end agentic AI systems using LangChain, LangGraph, and LangFuse, with a focus on industrializing Generative AI solutions for production customer deployments
- Build and maintain scalable backend services with FastAPI, containerized via Docker and Docker Compose, integrating vector databases (Weaviate) and PostgreSQL for robust data persistence and pipeline management
- Design and implement data engineering pipelines for context management and retrieval, including database architecture and data flow across multi-agent systems
- Drive concept creation for context management strategies and multi-agent architectures; act as a go-to person in AI agents, AgentOps, and LLM integration across major APIs (OpenAI, Anthropic, and others)

Fraunhofer IIS

Working Student – AI & Machine Learning

Tennenlohe, Bavaria
October 2023 – November 2025

- Contributed to machine learning pipelines with a focus on computer vision across projects such as SyNaKI and GAIA
- Implemented a distributed AI tool using evolutionary algorithms, tailored to application-specific requirements
- Actively participated in preparing funding applications to support AI research and development within Fraunhofer IIS
- Optimized vision models for deployment on embedded systems
- Recognized as a dependable and proactive team member within the AI team, supporting both research and practical implementation

DIT Lab for Digitalization of AI in Electrical Engineering

Research Assistant

Deggendorf, Bavaria
March 2022 – March 2023

- Led data acquisition for sensor-based DC-DC converters to enable predictive maintenance through behavioral monitoring
- Performed time series analysis on collected sensor data to extract patterns and detect anomalies
- Built, trained, optimized, and deployed statistical models and neural networks on resource-constrained devices (TinyML)
- Conducted Bachelor's thesis on TinyML as part of research at the lab

T-Systems Multimedia Solutions GmbH

Internship – IoT Edge Computing & Machine Learning (AWS)

Dresden, Saxony
October 2021 – February 2022

- Independently designed and deployed scalable cloud services and end-to-end data engineering pipelines using the AWS Cloud Development Kit (CDK), including database management and data visualization dashboards
- Contributed to predictive maintenance projects for automated material handling systems and ultrapure water valve condition monitoring
- Gained practical experience with SCRUM and CI/CD methodologies in an agile environment

DIT Project Lab for Hardware-Related Digitization

Research Assistant – Embedded Systems

Deggendorf, Bavaria
October 2020 – July 2021

- Developed a GUI application enabling communication with various sensors via UART, SPI, and I²C protocols using the PICKit Serial Analyzer
- Contributed to bridging hardware and software integration in embedded systems projects

Education

Friedrich-Alexander-University (FAU)

M.Sc. Artificial Intelligence

Erlangen, Bavaria
April 2023 – March 2026

- German grade: 1.3 | Focus: Advanced Deep Learning, Pattern Recognition, Statistical Learning, Diffusion and Attention Models
- Master Thesis: K-space Latent Diffusion for Accelerated MRI (IdeaLab) – Grade: 1.3

Deggendorf Institute of Technology (DIT)

B.Eng. Applied Computer Science

Deggendorf, Bavaria
October 2019 – March 2023

- German grade: 1.4 | Focus: Embedded Systems
- Bachelor Thesis: Generation, Optimization, and Evaluation of ML/DL Models for Tiny Machine Learning (TinyML) – Grade: 1.3

University of Southern Denmark (SDU)

Summer School – Deep Learning

Odense, Denmark
August 2022

- Completed a hands-on project classifying healthy vs. pneumonia-affected lungs using convolutional neural networks

Research Projects

SMMetrDE – Segmentation-guided Monocular Metric Depth Estimation

- Designed a framework combining semantic segmentation with monocular depth estimation to derive object-wise metric depth
- Integrated and evaluated multiple state-of-the-art depth models (MonoDepth2, MiDaS, DPT, ZoeDepth) and optimized them for metric prediction
- Conducted model performance evaluations using RMSE, AbsRel, and inference time across KITTI dataset subsets
- Gained practical experience with transformer-based vision models (ViT, BEiT, Swin) and their applications in depth estimation

AI Agent for Kalah – Tournament Challenge

Competitive game-playing agent using search and decision-making algorithms

- Applied minimax search, alpha-beta pruning, and heuristic evaluation to optimize gameplay decisions under strict time constraints

Deep Learning from Scratch

Core deep learning architectures implemented from scratch in Python

- Implemented forward/backward passes for FNNs, CNNs, and RNNs, alongside custom optimizers (SGD, Adam), activation functions, and loss functions

Technical Skills

AI / ML Frameworks: LangChain, LangGraph, LangFuse, PyTorch, PyTorch Lightning, TensorFlow, Keras, Hugging Face, OpenCV

Agent & LLM Technologies: Multi-agent systems, AgentOps, RAG pipelines, Weaviate (vector DB), OpenAI / Anthropic / Gemini APIs

Infrastructure & DevOps: FastAPI, Docker, Docker Compose, AWS (CDK), Git, GitHub, CI/CD, Jira, PostgreSQL

Programming Languages: Python, C, C++, SQL, LaTeX

Hardware & Embedded: Raspberry Pi, Arduino, STM32, SPI, I2C, UART, MATLAB

Languages: German (Native), English (C1), French (B1)

Awards & Certificates

Deutschlandstipendium 2021 – Sponsored by Vector Informatik GmbH

Honorary Award 2019 – Head of Technical Team, Tassilo Gymnasium

Python for Data Science, ML & AI – Udemy, 2022